

Senior Ontology - Smart Route & User Management API

Unit test specification: User Registration (1.0)

Scope note: Backend tests target server.py (FastAPI). DFD processes 1.1–1.3 are mapped to API endpoints and mocked User Info store. Flutter UI validation (field format on device) is out of scope for these Python unit tests.

Test suite A — Process 1.0 User Registration

Test Case ID	TC-REG-1.0
Python module	UnitTest_1_0_User_Registration.py
Priority	High
Data store	User Info (Key.json / mocked read_users, save_users)

DFD sub-processes covered

1.1 Get user information — Five fields (firstname, lastname, email, phonenumber, password) are accepted by POST /signup via model UserRegister. Test: test_dfd_1_1_get_user_info_signup_payload (signup payload).

1.2 Check existing user — Duplicate email rejected before save. GET /check-email exposes existence check for UI. Tests: test_dfd_1_2_check_existing_user_duplicate, test_dfd_1_2_check_email_endpoint.

1.3 Record user information — On success, save_users persists the new row. Post-registration password update uses the same store. Test: test_dfd_1_3_record_and_post_registration_password (store length 1 after signup; change_password 401/404/success).

Step	Action	Expected Result	Status
1	POST /signup with full UserRegister	201, User created	Pass
2	POST /signup duplicate email	400, Email already registered	Pass
3	GET /check-email existing vs missing	exists true/false	Pass
4	POST change_password after signup (valid / invalid paths)	401 / 404 / Password updated	Pass

DFD / requirement	Unit test method
1.1–1.3 Registration flow	test_dfd_1_1_get_user_info_signup_payload, test_dfd_1_2_*
1.2 Duplicate + check-email	test_dfd_1_2_check_existing_user_duplicate, test_dfd_1_2_check_email_endpoint
1.3 Persist + password maintenance	test_dfd_1_3_record_and_post_registration_password

Coverage: run backend_tests/run_coverage.sh — UnitTest_1_0_User_Registration.py 100%. Generated 2026-04-06 10:53.